Lesson 7-7 · Session 1

Polygons and Perimeter on the Plane

Name _____ Date _____ Period _____

Maryland Standard 6.G.A.3 · Learning target — I can find the perimeter and area of a polygon drawn on the coordinate plane.
Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

Learn from a model

REMEMBER

- You can plot a polygon's vertices on a coordinate plane, then find its side lengths using the distance between points.
- Perimeter = the sum of all the side lengths.
- Watch out — a side that crosses zero needs its two absolute values added, not subtracted.
- Check it — sketch the polygon and label each side before you add or multiply.

WORKED EXAMPLE

Finding the Perimeter of a Rectangle from Its Vertices A rectangle has vertices at (−8.5, 7.5), (5.5, 7.5), (5.5, −3.5), and (−8.5, −3.5). Find its perimeter.

Step 1 Plot the four vertices and connect them in order to draw the rectangle.

Step 2 Top side: the y-coordinates match, so use the x-values: |−8.5| + |5.5| = 14

Step 3 Left side: the x-coordinates match, so use the y-values: |7.5| + |−3.5| = 11

Step 4 Perimeter adds all four sides: 14 + 11 + 14 + 11 = 50

Answer The rectangle measures 14 units by 11 units, so its perimeter is 50 units.

Practice

1. A rectangle has vertices (0,0), (0,4), (6,4), (6,0). Find the length of each side, then the perimeter.

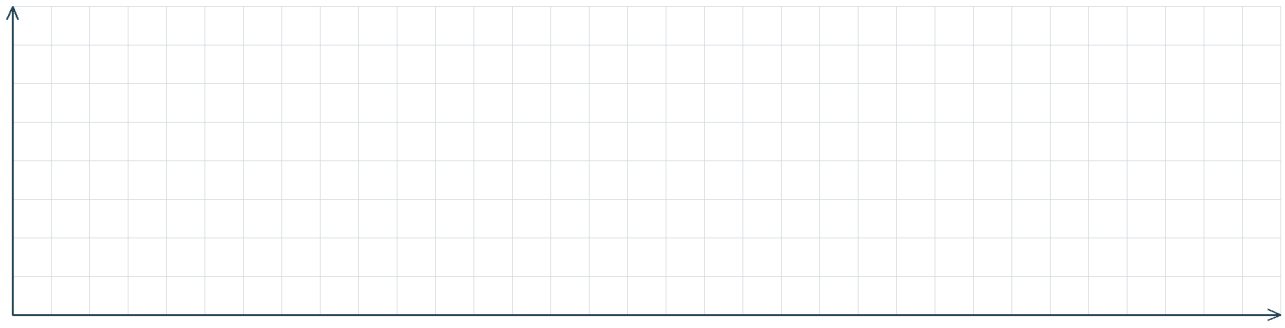
Answer _____

SHOW YOUR THINKING

2. A right triangle has vertices (0,0), (0,5), (3,0). Find its two leg lengths using the grid.

Answer _____

DRAW YOUR GRAPH / LABEL BOTH AXES AND THE SCALE



3. Explain how you find a horizontal or vertical side length using coordinates. Sentence Frame: "I know this because..."

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

4. Write each answer on the line.

a. Distance between (−4, 3) and (7, 3): _____
b. Distance between (2, 9) and (2, −5): _____
c. Perimeter of a rectangle 8 units by 5 units: _____
d. Perimeter of a square with side 9 units: _____
e. Area of a rectangle 6 units by 7 units: _____

Answer _____

SHOW YOUR THINKING

5. Sandbox. A sandbox has corners at (−5, 2), (7, 2), (7, −6), and (−5, −6). Each unit is one foot. Find both side lengths.

Answer _____

SHOW YOUR THINKING

6. Poster Border. A poster has corners at (−3, 10), (9, 10), (9, −4), and (−3, −4). Each unit is one inch. Find the perimeter of the poster.

Answer _____

SHOW YOUR THINKING

7. Fourth Corner. Three corners of a rectangle sit at (−8, 6), (4, 6), and (4, −5). Find the fourth corner, then find the perimeter in grid units.

Answer _____

SHOW YOUR THINKING

8. For a side joining (−9, 2) and (6, 2), Bri wrote the length as 3 units, because nine minus six is three. Explain the mistake, then show the correct length.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

9. A rectangle has vertices at (−4, 3), (8, 3), (8, −5), and (−4, −5). Find its perimeter.

Answer _____ units

SHOW YOUR THINKING

10. For the same rectangle, find the area.

Answer _____ square units

SHOW YOUR THINKING

11. Find the Sides. A rectangle has vertices at (2, 9), (11, 9), (11, 1), and (2, 1). Find both side lengths.

Answer _____ both sides

SHOW YOUR THINKING

12. Find the Mistake. For a side from (−5, 2) to (4, 2), Jo wrote 1 unit. Explain the mistake.

Answer _____ the mistake and the length

EXPLAIN WITH MATHEMATICAL EVIDENCE

Spiral review

13. From Lesson 7-6 Session 1. Find the distance from (6, −3) to (6, 5).

Answer _____

SHOW YOUR THINKING

14. From Lesson 6-6 Session 1. Simplify 8k + 1 + 2k.

Answer _____

SHOW YOUR THINKING

15. From Lesson 4-4 Session 1. Find 40% of 175.

Answer _____

SHOW YOUR THINKING

Test practice

16. A rectangle with sides 11 and 6 has a perimeter of —

□ A. 66	□ B. 34
□ C. 17	□ D. 22

17. A rectangle with sides 11 and 6 has an area of —

□ A. 17	□ B. 34
□ C. 66	□ D. 132

18. The distance from (0, −4) to (0, 6) is —

□ A. 24	□ B. 2
□ C. −10	□ D. 10

Check and revise

CHECK YOUR MATHEMATICS	HOW DID IT GO?
Use the scale and count every side once. Every part answered / Work shown / Units labeled when needed	I can do this: Independently / With a hint / With help Question to revisit: _____

One correction or question I want to discuss: _____

Check signs, scale, and labels. Use a number line or coordinate grid to verify one response.